Analysis of Dog Registration Compliance & Incentives

Identifying Late Payment Patterns and Incentive Effectiveness

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1. Executive Summary

This report analyses dog registration data from **Hamilton City Council between 2020 and 2024** to understand registration behaviour, late payment patterns, and participation in council discount programs.

Using **SQL Server for data preparation and Power BI for analysis and visualisation**, the project identifies trends that may help the council improve compliance and target communication more effectively.

Key Findings

- **Seasonal registration behaviour**
  Dog registrations follow a clear seasonal pattern. Most payments occur in **July**, which corresponds with the annual registration renewal period. Late payment penalties increase shortly after this peak, indicating some owners delay registration beyond the due date.
- **Variation across suburbs**
  Late payment rates differ across suburbs. Areas such as Enderley, Bader, and Melville show higher late payment rates, while suburbs such as Harrowfield and Huntington show lower rates.
- **Uneven participation in discount programs**
  Participation in available discounts (such as fenced property, neutering, obedience training, and permanent identification) varies widely between suburbs. Some areas show relatively high participation, while others rarely use these incentives.
- **Dog population concentration**
  Dog ownership is concentrated in several suburbs, including Nawton, Frankton, Hamilton East, Dinsdale, and Chartwell. Because these areas have the highest number of registered dogs, improvements in compliance here could have the greatest impact overall.

Recommendations

Based on the analysis, the following actions may help improve registration compliance:

- Send registration reminders **before the July renewal period**
- Focus outreach efforts in suburbs with **higher late payment rates**
- Increase awareness of **available discount programs**
- Prioritise communication efforts in **high-population suburbs**
- Continue monitoring **data quality** to support accurate reporting

This analysis provides a data-driven overview of registration behaviour and highlights opportunities for targeted communication and policy improvements.

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2. Business Context & Objectives

Problem Statement

Dog registration is a regulatory requirement managed by Hamilton City Council. Ensuring timely registration helps maintain accurate records and supports responsible pet ownership.
However, some dog owners register late or do not take advantage of available incentives. Understanding where and when these behaviours occur can help the council improve compliance and design more effective communication strategies.

Project Objectives

The objectives of this analysis are to:

- Identify suburbs with **higher rates of late dog registration**
- Analyse geographic patterns in **discount program participation**
- Explore possible relationships between **late payments and incentive uptake**
- Provide **interactive Power BI dashboards** that allow stakeholders to explore:
    - suburb-level trends
    - time-based registration patterns
    - differences between registration categories

Stakeholders

This analysis is intended to support several council teams:

- **Council Managers**
  Use insights to inform policy decisions and evaluate existing programs.
- **Dog Registration Team**
  Identify areas that may require additional operational support or outreach.
- **Communications Team**
  Design targeted campaigns to encourage timely registration and increase awareness of incentives.

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3. Data Overview & Methodology

This section describes the datasets used in the analysis, how they were prepared, and the analytical approach used to generate insights.

3.1 Dataset Description

The analysis uses publicly available datasets published through the **Waikato Open Data Co-Lab**, which provides open data from Hamilton City Council. The analysis is intended to demonstrate data analytics techniques and does not represent official work conducted for Hamilton City Council.

These datasets are licensed under the Creative Commons Attribution 4.0 New Zealand License, which allows reuse and analysis with appropriate attribution.

The data was accessed on 16 February 2026 via the Waikato Open Data Portal.

Table	Records	Key Purpose
DimDog (Dimension Dog)	82,149	Demographics & geography
FactRegistration (Dog Registration)	323,196	Registration & fees
FactTransaction (Dog Transaction)	1,550,635	Payment behaviour & discounts

Key fields used in the analysis include:

- **Dog_Number** - unique identifier for each dog
- **Registration_Key** - unique identifier for each registration record
- **Kept_At_Suburb** - suburb where the dog is located
- **Fee_Code_Description** - describes owner categories, useful for comparing compliance versus default owners: Category 1 = compliant owners (offence-free 24 months, eligible for reduced fees), Category 2 = default owners (pay full fee) Hamilton City Council - Renew a Dog Registration.
- **Transaction_Type** - type of transaction (e.g. prompt payment, late penalty, discount)

Only fields relevant to the analysis were retained to simplify the data model and improve performance.

3.2 Data Structure & Relationships

A star schema data model was implemented in Power BI to organise the data efficiently for analysis. The model separates descriptive information (dog demographics) from transactional data (registrations and payments), as illustrated in Figure 1.

Star Schema Diagram:

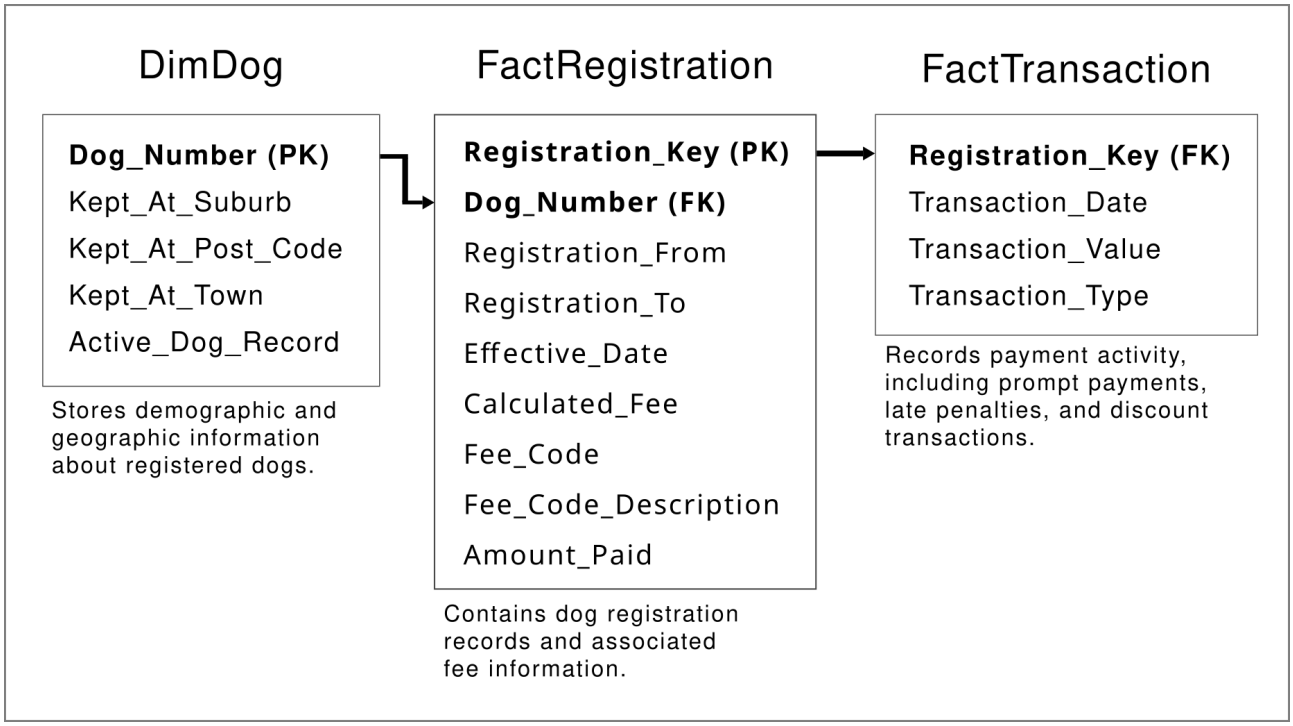

Figure 1: Star Schema for Hamilton Dog Registration Analysis

Relationships:

- **DimDog** → **FactRegistration** via Dog_Number
- **FactRegistration** → **FactTransaction** via Registration_Key

This structure supports efficient aggregation and enables analysis of payment behaviour across locations and time periods.

3.3 Data Preparation & Cleaning

Data preparation was performed using SQL Server Management Studio and Power BI.

Key steps included:

1. Data filtering
The transaction dataset was filtered to include only records related to animal registration payments, which represent the majority of transactions in the dataset.

2. Handling missing data
Missing values were reviewed to understand their potential impact:

- Approximately 2.5% of suburb values were missing
- Around 12-13% of registrations were missing payment or effective date information

Records with missing suburb information were excluded from suburb-level analysis.

3. Removing duplicates
Duplicate registration records were identified and removed using SQL queries to ensure each registration was counted only once.

4. Data validation
Several checks were performed to identify potential issues:

- registrations with invalid date ranges
- negative or unusually large payment values
- inconsistent suburb names

Standardising suburb names ensured that geographic analysis remained accurate.

After cleaning and validation, the prepared dataset was imported into Power BI for modelling and visualisation.

3.4 Exploratory Data Analysis (EDA)

Exploratory data analysis was conducted to understand the structure of the dataset and identify initial patterns in registration behaviour, late payments, and participation in incentive programs.

Key areas explored included:

- **Dog population distribution**
  Total registered dogs were analysed by suburb to understand where dog ownership is concentrated across Hamilton, Waikato.
- **Late payment behaviour**
  Late payment penalties were reviewed across multiple years to identify recurring patterns and possible geographic differences.
- **Discount participation**
  Transactions related to incentive programs (fencing, neutering, obedience training, and permanent identification) were examined to understand participation levels.
- **Geographic and time-based patterns**
  Suburb-level metrics and monthly transaction trends were analysed to detect seasonal behaviour and spatial variation.

The insights from this stage helped define the key metrics and visualisations used in the final dashboards.

Analytical Summary
The EDA established baseline metrics for dashboards, including:

- Total dog registrations per suburb
- Late payment rates by suburb and registration category
- Discount uptake percentages by incentive type
- Monthly trends for prompt and late payments

Detailed SQL scripts, data quality checks, and transformations are included in Appendix B. The resulting dataset was imported into Power BI, where a star schema and DAX measures supported interactive dashboards for exploring compliance, incentive uptake, and geographic variation.

3.5 Analytical Approach

The analysis focused on understanding dog registration compliance, late payment behaviour, and uptake of discount incentives. Data was prepared using SQL, modelled in Power Query, and analysed in Power BI using DAX measures. The workflow followed four main components:

1. **Geographic Analysis**
   Suburb-level metrics identified areas with higher late payment rates and discount uptake. SQL queries aggregated dog populations, late payment counts, and incentive participation, while Power Query cleaned and standardised suburb groupings. These insights help the Dog Registration and Communications teams prioritise targeted outreach.

2. **Discount Uptake Analysis**
   Participation in incentive programs was analysed across suburbs and registration categories. SQL identified transactions linked to discounts, including cases applied after late payments. Power Query grouped fee categories for consistency, and DAX calculated discount uptake percentages for dashboard KPIs. Council Managers can use this to assess program effectiveness and funding allocation.

3. **Late Payment Analysis**
   Patterns of late payments were examined over time and by registration category. SQL extracted counts of late penalties and prompt payments, while DAX measures quantified key metrics such as late registration rates. These results support all stakeholders in monitoring compliance and identifying recurring behaviours that may require intervention.

4. **Temporal Trend Analysis**
   Monthly and yearly trends in registrations, payments, and discount uptake were analysed for 2020-2024. Time-series SQL queries provided transaction counts, and DAX enabled visualisation of prompt versus late payments and total transaction volumes. This allows teams to detect seasonal or event-driven patterns and plan interventions accordingly.

Summary:
After cleaning and validating the dataset, it was imported into Power BI using a star schema. Interactive dashboards enable stakeholders to explore compliance patterns, discount uptake, and geographic variation, supporting data-driven operational and policy decisions.

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4. Analytical Findings

This section presents an analysis of dog registration data between 2020 and 2024. The objective is to understand registration behaviour, compliance trends, geographic patterns, and the adoption of discount programs. All analyses were performed using SQL queries and visualised in Power BI. Observations account for missing data, incomplete transactions, and seasonal effects.

Note: This analysis is exploratory and descriptive; figures reflect recorded transactions and registrations. Some records have missing payment amounts or effective dates, which may slightly affect counts or rates.

4.1 Registration Activity Shows Strong Annual Peaks

Dog registration activity follows a consistent seasonal pattern.
As shown in Figure 2, the transaction volumes increase sharply each **July**, when annual registrations are due.

Most prompt payments occur during this period, while late payment penalties tend to appear shortly afterwards.

This pattern indicates that the majority of dog owners comply during the renewal period, but a smaller portion register after the due date.

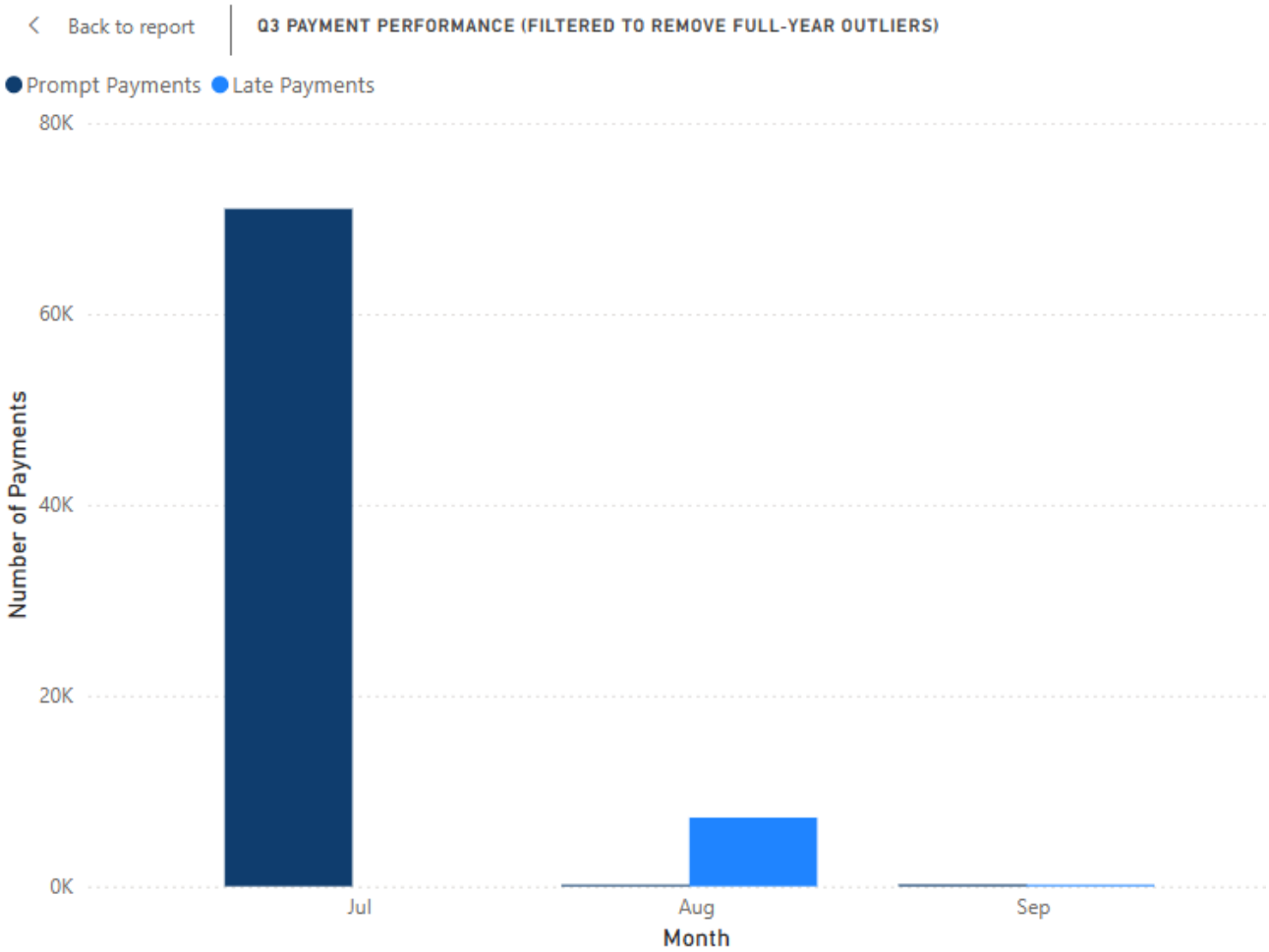

Figure 2: Prompt vs Late Dog Registration Payments - Q3 Performance

4.2 Late Payment Behaviour Varies by Suburb

Late payment rates vary considerably between suburbs.

Suburbs with higher late payment rates include Enderley (20%), Avalon (20%), Bader (20%), Temple View (18%), and Melvile (17%), as illustrated in Figure 3.

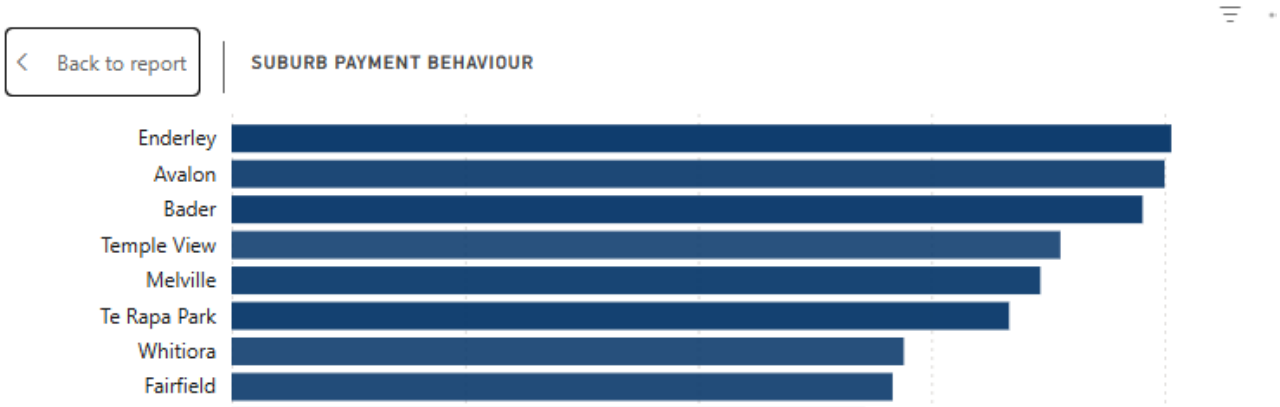

Figure 3: Highest Late Payment Rate by Suburb

Lower rates are seen in suburbs like Huntington (6%), Te Rapa (5%) and Harrowfield (5%), as illustrated in Figure 4.

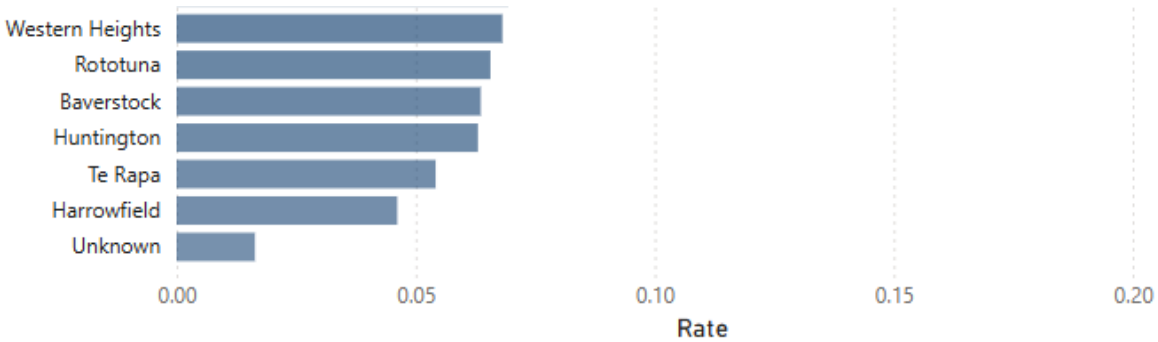

Figure 4: Lowest Late Payment Rate by Suburb

On the map, late payment behaviour shows the full range, with a slight concentration on the western side of Hamilton, as illustrated in Figure 5.

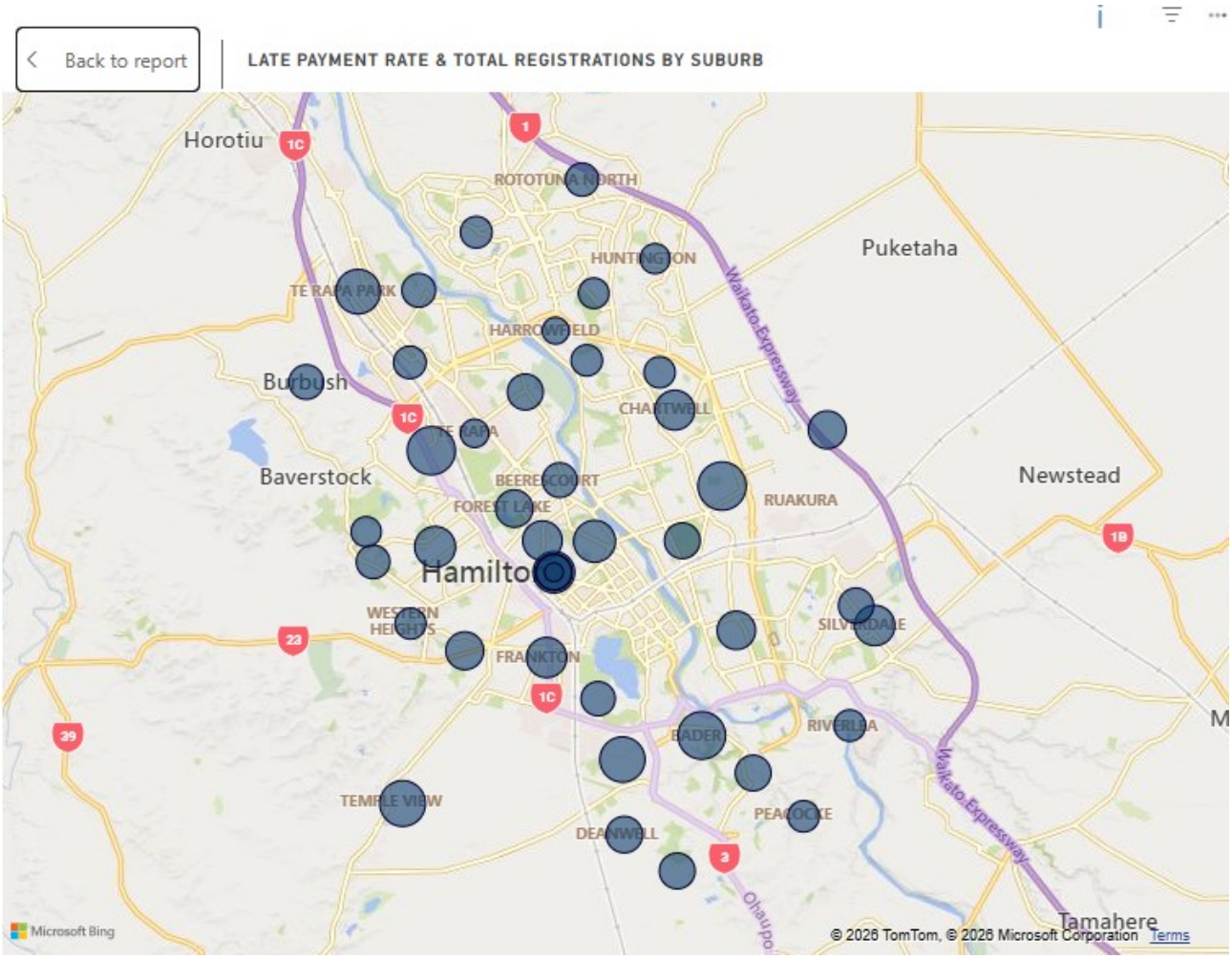

Figure 5: Geographic Distribution of Late Payments

4.3 Discount Program Adoption

Hamilton City Council offers several discounts for responsible dog ownership, including incentives related to: Fenced properties, Neutered dogs, Obedience training, Permanent identification. Participation in these programs varies across suburbs.

Examples of participation areas:

Highest Uptake Suburb	Higher Discount Rate	Lowest Uptake Suburb	Lower Discount Rate
Te Rapa Park	45.8%	Huntington	10.9%
Rural Hamilton 1	44.8%	Rototuna North	11.9%
Peacocke	37.6%	Baverstock	1.5%

4.4 Relationship Between Late Payments and Discounts

An exploratory comparison was conducted to examine whether suburbs with higher participation in discount programs also show lower late payment rates.

The Power BI dashboard was used to compare suburb-level **late payment rates and discount participation rates**. While some areas with higher discount participation appear to have lower late payment rates, the overall analysis did **not show a consistent or measurable correlation** between the two variables.

This suggests that participation in discount programs does not necessarily correspond with earlier registration behaviour. Additional analysis, including statistical testing or owner-level behavioural data, would be required to determine whether these incentives influence payment timing.

4.5 Dog Population Distribution

Dog ownership is unevenly distributed across Hamilton.

Suburbs with the largest dog populations include:

Suburb	Registered Dogs
Nawton	7,735
Frankton	7,705
Hamilton East	5,926
Dinsdale	5,380
Chartwell	5,241

Because these areas contain the largest number of dogs, improvements in compliance within these suburbs could significantly affect overall registration rates.

4.6 Data Quality Considerations

While overall data quality is good, several limitations were identified:

- approximately 13% of registrations have missing payment values
- around 12% have missing effective dates
- a small number of registrations have no associated transaction record

These issues do not invalidate the analysis but should be considered when interpreting results.

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5. Key Insights

The analysis highlights several important patterns in dog registration behaviour.

1. Registration activity is highly seasonal
Most dog registrations occur in July, making this the most important period for communication campaigns and operational planning.

2. Late payments vary across suburbs
Some suburbs consistently show higher late payment rates, suggesting that targeted outreach may be more effective than city-wide messaging.

3. Discount participation differs significantly between areas
While some suburbs actively participate in discount programs, others rarely use them, indicating potential gaps in awareness.

4. High-population suburbs have the greatest impact
Because suburbs such as Nawton and Frankton contain large numbers of dogs, improving compliance in these areas could significantly improve overall results.

5. Data quality monitoring is important
Missing payment and transaction data should continue to be monitored to maintain accurate reporting.

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6. Recommendations

Based on the analysis, several actions may help improve registration compliance and participation in incentive programs.

1. Send Reminders Before the Registration Peak

- Action: Launch reminder campaigns in June before the annual July registration period.
- Rationale: Early communication may encourage owners to register before the deadline and reduce late payments.

2. Focus Outreach on High Late-Payment Suburbs

- Action: Prioritise outreach in suburbs with higher late payment rates such as Enderley, Bader, and Melville.
- Rationale: Targeted communication can address areas where delayed registration is more common.

3. Promote Awareness of Discount Programs

- Action: Increase promotion of available incentives through council communication channels.
- Rationale: Greater awareness may encourage responsible ownership and improve registration compliance.

4. Prioritise High-Population Areas

- Action: Focus communication efforts in suburbs with the largest dog populations.
- Rationale: Improvements in these areas will have the greatest impact on overall compliance.

5. Continue Monitoring Data Quality

- Action: Regularly review registration and transaction records for missing or inconsistent data.
- Rationale: Reliable data supports better reporting and decision-making.

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7. Opportunities for Further Investigation

While this project focuses on descriptive analysis of registration behaviour, several areas could be explored further to provide deeper insights.

Possible future investigations include:

- **Detailed monthly trend analysis**
  Exploring more granular monthly or weekly patterns may reveal behavioural changes leading up to the July registration deadline.
- **Demographic and socioeconomic factors**
  Integrating external datasets such as household income, property types, or population density could help explain differences in registration behaviour across suburbs.
- **Evaluation of individual discount programs**
  Analysing which specific incentives (e.g., neutering or fencing) have the highest participation may help improve program design.
- **Automated data quality monitoring**
  Implementing automated checks for missing or inconsistent records would improve the reliability of ongoing reporting.

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8. Limitations & Assumptions

Several limitations should be considered when interpreting the results of this analysis.

- **Missing data**
  Some registration records contain missing payment values (~13%) or missing effective dates (~12%). While relatively small, these gaps may slightly affect calculated rates.
- **Transaction timing**
  Because most payments occur in July, monthly transaction charts outside this period may show very low activity.
- **Discount classification**
  Discounts were identified using transaction descriptions. Some records may be miscoded or incomplete.
- **Descriptive scope**
  The analysis identifies patterns in historical data but does not establish causal relationships between incentives and payment behaviour.

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9. Conclusion

This project analysed dog registration data from **2020-2024** to identify patterns in payment behaviour, discount participation, and geographic variation.

The analysis found that:

- registrations peak strongly in **July**
- late payment behaviour varies across suburbs
- participation in discount programs differs widely
- several suburbs contain a large share of the city's dog population

These insights highlight opportunities for **targeted communication, improved awareness of incentives, and better operational planning**.

Overall, the analysis demonstrates how council data can be used to support **evidence-based decision making and more effective service delivery.**

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10. References

Hamilton City Council – Renew a Dog Registration
https://hamilton.govt.nz/do-it-online/pay-it/renew-a-dog-registration

Waikato Open Data Co-Lab – Dog Dataset (Hamilton City Council)
https://data-waikatolass.opendata.arcgis.com/datasets/hcc::dog/about

Waikato Open Data Co-Lab – Dog Registration Dataset
https://data-waikatolass.opendata.arcgis.com/datasets/hcc::dog-registration/about

Waikato Open Data Co-Lab – Dog Transaction Dataset
https://data-waikatolass.opendata.arcgis.com/datasets/hcc::dog-transaction/about

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11. Appendices

The appendices show representative SQL queries, Power BI transformations, and DAX measures used in this analysis. Not all scripts or charts are included here. The full project files, including complete SQL scripts, Power BI dashboards, and documentation, are available in the project GitHub repository

Link: https://github.com/gabywu/dog-registration-analysis

A. SQL Server - Data Cleaning

Listing A.1. Create FactTransaction Table with Animal Registration Filter

```
SELECT *
INTO dbo.FactTransaction
FROM dbo.Dog_Transaction_Stage
WHERE Type_Of_Account = 'Animal Registration';
```

Creates a filtered FactTransaction table containing only "Animal Registration" accounts. Ensures subsequent analyses focus on relevant transactions linked to registered dogs.

Listing A.2. Audit Missing Values in DimDog Key Fields

```
DECLARE @TableName SYSNAME = 'DimDog';
DECLARE @SchemaName SYSNAME = 'dbo';
DECLARE @SQL NVARCHAR(MAX) = N'';
SELECT @SQL = STRING_AGG(
    'SELECT ''' + COLUMN_NAME + ''' AS ColumnName, ' +
    'SUM(CASE WHEN [' + COLUMN_NAME + '] IS NULL THEN 1 ELSE 0 END) AS Nulls, '
+
    'CAST(SUM(CASE WHEN [' + COLUMN_NAME + '] IS NULL THEN 1 ELSE 0 END) * 100.0
/ COUNT(*) AS DECIMAL(8,2)) AS NullPct ' +
    'FROM ' + QUOTENAME(@SchemaName) + '.' + QUOTENAME(@TableName),
    CHAR(13) + 'UNION ALL' + CHAR(13)
)
FROM INFORMATION_SCHEMA.COLUMNS
WHERE TABLE_SCHEMA = @SchemaName
  AND TABLE_NAME = @TableName
  AND COLUMN_NAME IN (
      'Dog_Number','Kept_At_Suburb',
      'Kept_At_Post_Code','Kept_At_Town','Active_Dog_Record'
  );
EXEC sp_executesql @SQL;
```

Audits missing values in key DimDog fields. Helps identify geographic or demographic columns requiring cleaning or exclusion to maintain data quality.

Listing A.3. Remove Duplicate Records in FactRegistration

```
WITH Dedup AS (
SELECT *, ROW_NUMBER() OVER (
 PARTITION BY Registration_Key, hash_binary
 ORDER BY FID
) AS rn
 FROM dbo.FactRegistration
) DELETE FROM Dedup WHERE rn > 1;
```

Removes duplicate registrations using a row-number window function. Ensures only unique records per registration key and hash are retained for accurate analysis.

Listing A.4. Standardise Fee Code Descriptions in FactRegistration

```
UPDATE dbo.FactRegistration
SET Fee_Code_Description = 'Probationary & Dangerous'
WHERE Fee_Code_Description = 'Probat. & Dangerous';

UPDATE dbo.FactRegistration
SET Fee_Code_Description = 'Disability Assist Dog'
WHERE Fee_Code_Description = 'DisabilityAssistDog';
```

Ensures consistent naming for fee categories across registration records, simplifying reporting and analysis.

Listing A.5. Validate Registration Date Order

```
SELECT Registration_Key, Registration_From, Registration_To
FROM dbo.FactRegistration
WHERE Registration_From > Registration_To;
```

Identifies records where registration start dates are later than end dates. Ensures temporal consistency for trend and compliance analyses.

Listing A.6. Identify Outliers in Payments and Transactions

```
SELECT *
FROM FactRegistration
WHERE TRY_CAST(Amount_Paid AS DECIMAL(10,2)) < 0
   OR TRY_CAST(Amount_Paid AS DECIMAL(10,2)) > 500
ORDER BY TRY_CAST(Amount_Paid AS DECIMAL(10,2));

SELECT *
FROM FactRegistration
WHERE Calculated_Fee < 0
   OR Calculated_Fee > 500
ORDER BY Calculated_Fee;
```

Flags negative or unusually high payment values, ensuring metrics are not skewed by anomalies.

Listing A.7. Detect Suburb Anomalies

```
SELECT DISTINCT Kept_At_Suburb, COUNT(*) AS CountSuburb
FROM dbo.DimDog
WHERE Kept_At_Suburb LIKE 'Rural Hamilton%'
    OR Kept_At_Suburb IS NULL
GROUP BY Kept_At_Suburb
ORDER BY CountSuburb;
```

Identifies missing, placeholder, or unusually named suburbs. Supports accurate geographic and behavioral analyses.

B. SQL Server - Exploratory Data Analysis (EDA)

Dog Population by Suburb

```
SELECT Kept_At_Suburb, COUNT(*) AS Dog_Count
FROM dbo.DimDog
WHERE Kept_At_Suburb IS NOT NULL
GROUP BY Kept_At_Suburb
ORDER BY Dog_Count DESC;
```

Aggregates the number of registered dogs per suburb. Provides a baseline for geographic analysis and later calculation of rates such as late payments or discount uptake.

Listing B.1. Late Payment Transactions Over Time

```
SELECT
 YEAR(Transaction_Date) AS Year,
 MONTH(Transaction_Date) AS Month,
 COUNT(*) AS LatePayment_Count
FROM dbo.FactTransaction
WHERE Transaction_Type = 'Late Payment Penalty Fee'
```

```
GROUP BY YEAR(Transaction_Date), MONTH(Transaction_Date)
ORDER BY Year, Month;
```

Analyses temporal patterns in late payment penalties, highlighting recurring non-compliance and seasonal trends in dog registration behaviour.

Listing B.2. Discounts Applied After Late Payment Penalties

```
WITH Late AS (
 SELECT Registration_Key, MIN(Transaction_Date) AS LateDate
 FROM dbo.FactTransaction
 WHERE Transaction_Type = 'Late Payment Penalty Fee'
 GROUP BY Registration_Key
),
Discounts AS (
 SELECT Registration_Key, Transaction_Date AS DiscountDate
 FROM dbo.FactTransaction
 WHERE Transaction_Type LIKE '%Fenced%'
 OR Transaction_Type LIKE '%Neuter%'
 OR Transaction_Type LIKE '%Obedience%'
 OR Transaction_Type LIKE '%Permanent%'
)
SELECT d.Registration_Key, d.DiscountDate, l.LateDate
FROM Discounts d
JOIN Late l
 ON d.Registration_Key = l.Registration_Key
WHERE d.DiscountDate > l.LateDate;
```

Identifies cases where discounts were applied after late payment penalties. Helps evaluate whether financial incentives can still be redeemed post non-compliance.

Listing B.3. Late Payment Rate by Suburb

```
WITH LatePayments AS (
 SELECT fr.Registration_Key, dd.Kept_At_Suburb, 1 AS IsLate
 FROM dbo.FactTransaction ft
 JOIN dbo.FactRegistration fr
 ON ft.Registration_Key = fr.Registration_Key
 JOIN dbo.DimDog dd
 ON fr.Dog_Number = dd.Dog_Number
 WHERE ft.Transaction_Type = 'Late Payment Penalty Fee'
),
Registrations AS (
 SELECT fr.Registration_Key, dd.Kept_At_Suburb
 FROM dbo.FactRegistration fr
 JOIN dbo.DimDog dd
 ON fr.Dog_Number = dd.Dog_Number
)
SELECT
 r.Kept_At_Suburb,
 COUNT(lp.IsLate) AS LatePayment_Count,
 COUNT(*) AS Total_Registrations,
 CAST(COUNT(lp.IsLate) AS FLOAT) / COUNT(*) AS LatePayment_Rate
FROM Registrations r
LEFT JOIN LatePayments lp
 ON r.Registration_Key = lp.Registration_Key
```

```
GROUP BY r.Kept_At_Suburb
ORDER BY LatePayment_Rate DESC;
```

Calculates the proportion of registrations with late payment penalties per suburb. Supports geographic insights for targeted interventions or awareness campaigns.

Listing B.4. Discount Uptake by Suburb

```
WITH Discounts AS (
SELECT fr.Registration_Key, dd.Kept_At_Suburb, 1 AS IsDiscount
    FROM dbo.FactTransaction ft
    JOIN dbo.FactRegistration fr
        ON ft.Registration_Key = fr.Registration_Key
    JOIN dbo.DimDog dd
        ON fr.Dog_Number = dd.Dog_Number
    WHERE ft.Transaction_Type LIKE '%Fenced%'
       OR ft.Transaction_Type LIKE '%Neuter%'
       OR ft.Transaction_Type LIKE '%Obedience%'
       OR ft.Transaction_Type LIKE '%Permanent%'
),
Registrations AS (
    SELECT fr.Registration_Key, dd.Kept_At_Suburb
    FROM dbo.FactRegistration fr
    JOIN dbo.DimDog dd
        ON fr.Dog_Number = dd.Dog_Number
    WHERE dd.Kept_At_Suburb IS NOT NULL
)
SELECT
    r.Kept_At_Suburb,
    COUNT(d.IsDiscount) AS Discount_Count,
    COUNT(*) AS Total_Registrations,
    CAST(COUNT(d.IsDiscount) AS FLOAT) / COUNT(*) AS Discount_Rate
FROM Registrations r
LEFT JOIN Discounts d
    ON r.Registration_Key = d.Registration_Key
GROUP BY r.Kept_At_Suburb
ORDER BY Discount_Rate DESC;
```

Measures financial incentive participation across suburbs. Helps stakeholders evaluate which incentive programmes are most effective.

Listing B.5. Data Quality Check – Missing Payment or Effective Date

```
SELECT
    SUM(CASE WHEN Amount_Paid IS NULL THEN 1 ELSE 0 END) AS Null_AmountPaid,
    SUM(CASE WHEN Effective_Date IS NULL THEN 1 ELSE 0 END) AS
Null_EffectiveDate,
    COUNT(*) AS Total_Registrations
FROM dbo.FactRegistration;
```

Assesses data completeness. Identifies registrations with missing payment information or missing effective dates, ensuring KPI calculations are based on valid data.

C. Power BI – Power Query / Advanced Editor

Listing C.1. Suburb Aggregation in DimDog

```
#"Added Custom2" =
Table.AddColumn(#"Removed Columns", "Suburb_Aggregated", each if
Text.Contains([Suburb], "Rural Hamilton") then "Rural Hamilton" else [Suburb])
```

Aggregates all rural Hamilton suburbs into a single "Rural Hamilton" label while leaving other suburbs unchanged. This helps simplify geographic analysis and mapping.

Listing C.2. Full Suburb Name for Mapping

```
#"Added Custom3" =
Table.AddColumn(#"Replaced Value2", "SuburbGroup", each [Suburb_Aggregated] & ",
Hamilton, New Zealand")
```

Creates a fully qualified suburb name including city and country for consistent labeling in Power BI maps and tooltips.

Listing C.3. Fee Category Grouping in FactRegistration

```
 #"Added Custom" =
Table.AddColumn(#"Changed Type1", "FeeCategoryGroup", each let
Cat1List = {"CATEGORY 1","PROBATIONARY OWNER","REGISTRATION FEE","DANGEROUS
CATEGORY 1"},
Cat2List = {"CATEGORY 2","DANGEROUS CATEGORY 2"},
FeeCategoryGroup = if List.Contains(Cat1List,
Text.Upper(Text.Trim([FeeCodeDesc]))) then "Category 1" else if
List.Contains(Cat2List, Text.Upper(Text.Trim([FeeCodeDesc]))) then "Category
2"else "Other" in FeeCategoryGroup)
```

Standardises registration fee categories by grouping multiple fee codes into "Category 1," "Category 2," or "Other." This ensures consistent aggregation for analysis of payment behaviour and incentive uptake.

D. Power BI - Dax Measure

Listing D.1. Unique Dogs

```
Distinct Dogs =
DISTINCTCOUNT(DimDog[Dog_Number])
```

Counts the number of unique dogs in the population.

Listing D.2. Total Dog Registrations

```
Total Registrations =
DISTINCTCOUNT(FactRegistration[Registration_Key])
```

Counts all unique dog registrations in the dataset.

Listing D.3. Late Registrations

```
Late Registrations =
```

```
CALCULATE(DISTINCTCOUNT(FactTransaction[Registration_Key]),
FactTransaction[IsLatePayment] = 1)
```

Counts registrations where a late payment was recorded.

Listing D.4. Late Registration Rate (%)

```
Late Registration Rate =
DIVIDE([Late Registrations], [Total Registrations], 0)
```

Calculates the proportion of registrations that incurred late payments.

Listing D.5. Discount Uptake (%)

```
Discount Uptake =
COALESCE(DIVIDE([Discount Count], [Total Registrations], 0), 0)
```

Measures the percentage of registrations receiving a discount, handling zero values.

Listing D.6. Prompt Payment Transactions

```
Prompt Payments =
CALCULATE (COUNTROWS (FactTransaction), FactTransaction[Type] = "Prompt
Payment")
```

Counts all transactions where payment was made on time.

Listing D.7. Late Payment Transactions

```
Late Payments =
CALCULATE(COUNTROWS(FactTransaction), FactTransaction[IsLatePayment] = 1)
```

Counts all transactions with a late payment penalty.

Listing D.8. Late Payment Rate

```
Late Payment Rate =
DIVIDE ([Late Payments], [Late Payments] + [Prompt Payments], 0)
```

Calculates the ratio of late payments to all completed payments.

Listing D.9. Total Transactions

```
Transaction Count =
COUNTROWS(FactTransaction)
```

Counts all transaction records in the dataset.